

Multiple Matroid Secretary Problem

Matías Ortiz José Soto
Departamento de Ingeniería Matemática, Universidad de Chile

Introduction

Matroid Secretary Problem

- Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid and $w : E \rightarrow \mathbb{R}_+$ an injective weight function chosen by an adversary.
- The elements of the matroid are presented to an online algorithm in uniformly random order.
- The weights are initially unknown and revealed sequentially when an element is presented.
- Upon observing $w(e)$ of an element $e \in E$, one must irrevocably decide whether to include it in the final solution ALG.
- At the end, ALG must be independent ($\text{ALG} \in \mathcal{I}$).
- The goal is to maximize the following notion of competitiveness:

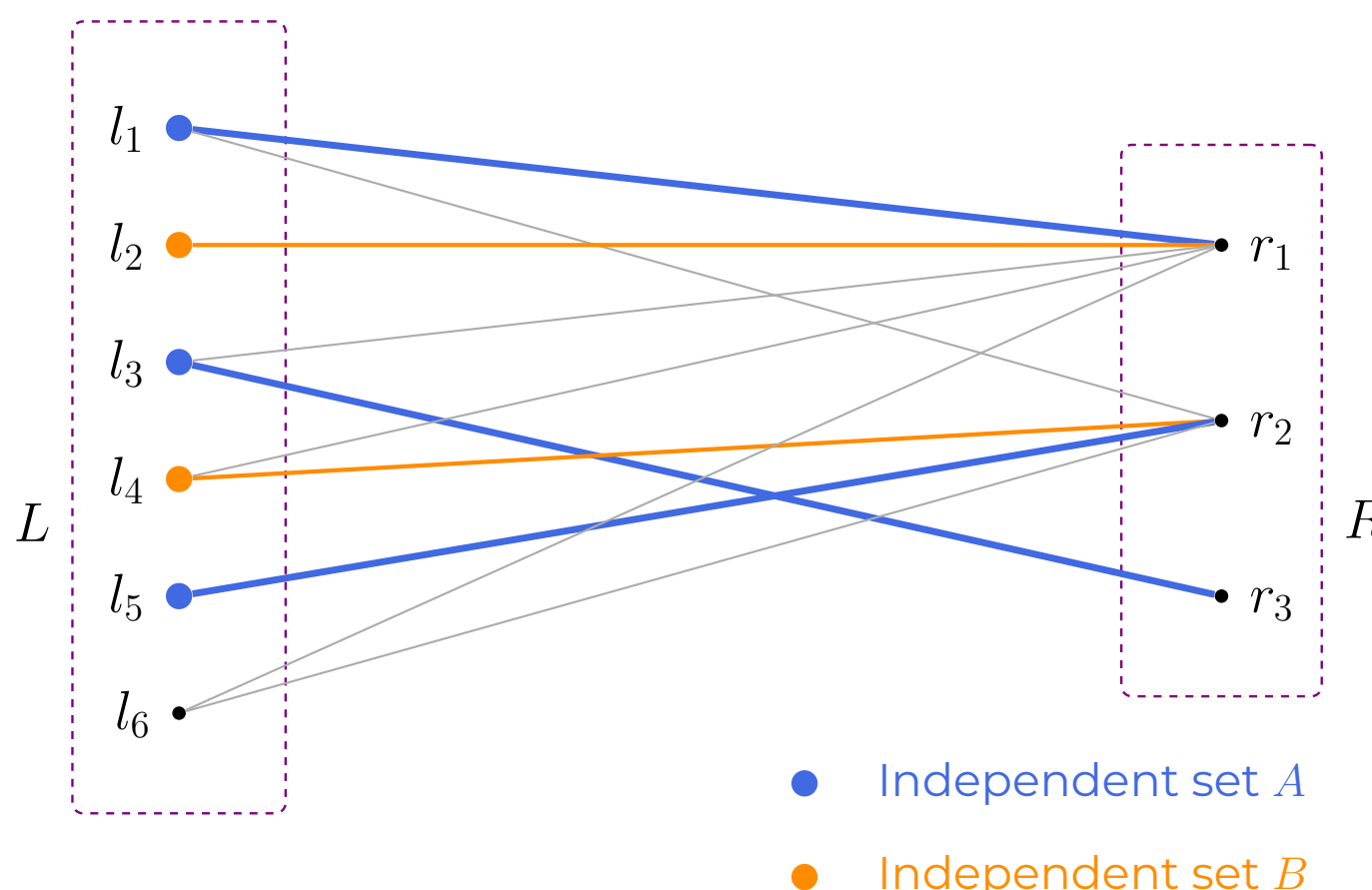
Definition (Probability Competitiveness).

We say that an online algorithm is α -probability competitive if for all $x \in \text{OPT}$:

$$\mathbb{P}(x \in \text{ALG}) \geq \alpha$$

Relaxation

In this research we define a relaxed version, of the *Matroid Secretary Problem*. Suppose the algorithm temporarily maintains two independent sets A and B , and finally returns $\text{OPT}(A \cup B)$. How much does this relaxation improve competitiveness? We motivate this with an example on the transversal matroid with two independent sets, this relaxation give much more “power” to an algorithm and the probability of selecting elements of OPT increases. But how much? What if we have more than two independent sets?



This generalization, which we call the *Multiple Matroid Secretary Problem*, is defined as follows:

Multiple Matroid Secretary Problem

- Let $\mathcal{M} = (E, \mathcal{I})$ be a matroid and $w : E \rightarrow \mathbb{R}_+$ an injective weight function chosen by an adversary.
- The elements of the matroid are presented to an online algorithm in uniformly random order. The weights are initially unknown and revealed sequentially when an element is presented.
- Unlike the classical MSP, the algorithm builds J independent sets $F_1, \dots, F_J \in \mathcal{I}$.
- Upon observing $w(e)$ of an element $e \in E$, one must irrevocably decide whether to include it in **one of the different independent sets F_i** , where $i \in \{1, \dots, J\}$.
- The final solution is $\text{ALG} = \text{OPT}(F_1 \cup \dots \cup F_J)$, the maximum-weight independent set contained in $\bigcup_{j=1}^J F_j$.
- The goal is to maximize the probability competitiveness.

Techniques

Continuous model

Instead of seeing n arriving elements in random order, we traverse the interval $[0, 1]$, where the arrival time of each element is a uniform random variable on that interval.

Our algorithms select elements of the matroid that satisfy certain requirements. The first condition is that the currently observed element must be an improving element, which we define below:

Improving Elements and Times

We say that $e_t = e \in E$ is an *improving* element if $e \in \text{OPT}(E_t)$. Where E_t is the set of elements revealed up to time t .

The remaining conditions refer to the number of previously accepted elements that “block” the current element. To control these notions and carry out all the corresponding analyses, we use the techniques and objects defined in Bérczi et al. [1].

Labeling. For each interval $[a, b]$ and for each improving time $t \in [a, b]$, we call *labeling* a bijection Λ_t that assigns to each element of $\text{OPT}(E_t)$ a unique label from $[r]$.

Improving Word. Let $0 \leq a < b \leq 1$, and let $\{e_k, e_{k-1}, \dots, e_1\}$ be the improving elements arriving in the interval $[a, b]$, with improving times $t_j = t_{e_j}$, ordered in decreasing order ($t_k < \dots < t_1$). An *improving word* in $[a, b]$ is $z = z_1 \dots z_k \in [r]^k$, where $z_j = z(t_j) = \Lambda_{t_j}(e_j)$ is the image of a labeling Λ_t on e_j . We say that $\Lambda = (\Lambda_t)_{t \in [a, b]}$ *generates*^a the word z in $[a, b]$.

^aAs indicated in the definition of labeling, in this random word only improving times contribute letters.



Example of an instance with sampling time p , and 4 improving elements arriving in $[p, 1]$ with labels assigned upon arrival, forming the random word $z = abcd$.

Proposition (Bérczi et al. Lemma 4 [1]). Let z be the random variable representing the *improving* word produced in the interval $[a, b]$, and let $|z|$ denote its length, with $\lambda = r \ln(b/a)$. Let $S, T \subseteq [r]$ be two disjoint sets of labels. We denote by z_S (respectively z_T) the subsequence of z containing only elements in S (respectively in T). Finally, let $\lambda_S = |S| \ln(b/a)$.

- For every $k \in \mathbb{N}$, $\mathbb{P}(|z| = k) = \mathbb{P}(N[a, b] = k) = \frac{\lambda^k e^{-\lambda}}{k!}$.
- Conditional on $|z| = k$, z is a uniformly random word in $[r]^k$.
- For every $k \in \mathbb{N}$, $\mathbb{P}(|z_S| = k) = \mathbb{P}(N_S[a, b] = k) = \frac{\lambda_S^k e^{-\lambda_S}}{k!}$.
- Conditional on $|z_S| = k$, z_S is a uniformly random word in S^k .
- z_S and z_T are independent.

Finally, our analyses reduce to constructing specific languages $\mathcal{L}$ for each matroid, which, given an element in the optimum ($e \in \text{OPT}$) and z its associated random word, verify that:

$$\mathbb{P}(e \in \text{ALG}) = \mathbb{P}(z \in \mathcal{L})$$

Greedy Type Algorithms in Transversal

Threshold Greedy Algorithm

Fix a sampling time $p \in [0, 1]$ and, after it, use a *cascading greedy* strategy: each improving element is inserted into the first independent set F_1 if $F_1 + e$ remains independent; otherwise, it is attempted in the second independent set, and so on up to F_J .

Multiple Threshold Algorithm

Another algorithm works in a less greedy manner: instead of a single sampling time, it uses J different thresholds ($p_1 \leq p_2 \leq \dots \leq p_J$), one for each independent set. It then applies the same cascading strategy: trying to insert each improving element into F_1 , if not possible then into F_2 , and so on up to F_J . The key difference is that now an element may only be introduced into the independent set F_i after time p_i . This algorithm was motivated by the optimal algorithm of Gilbert and Mosteller for the $(J, 1)$ -Secretary Problem [2].

Results

Transversal Matroid

Theorem:

For any $J \in \mathbb{N}$, the optimal algorithm for the $(J, 1)$ -Secretary Problem induces an algorithm with the same competitiveness in the Multiple MSP with J independent sets on the Transversal Matroid.

J	Probability Competitiveness	
	Threshold Greedy	Multiple Threshold
J=1	0.3679	0.3679
J=2	0.5869	0.5910
J=3	0.7260	0.7321
J=4	0.8167	0.8231
J=5	0.8767	0.8825
J=6	0.9167	0.9216
J=7	0.9444	0.9475

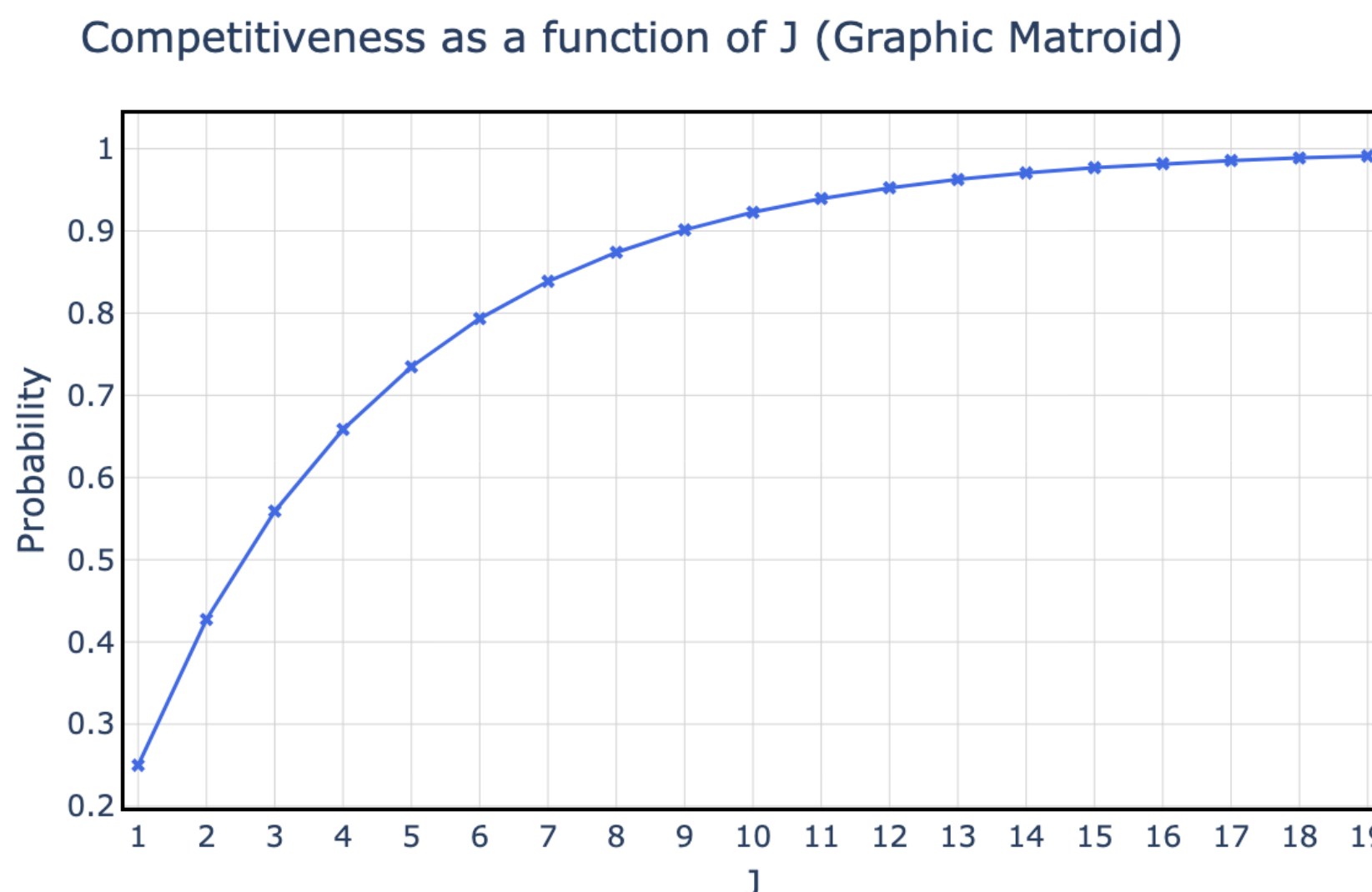
Corollary:

The Multiple Threshold Algorithm for J independent sets in the transversal matroid is optimal.

Graphic Matroid

Theorem:

There exists an algorithm for the Multiple Matroid Secretary Problem in the graphic matroid with two independent sets, achieving 0.4273 probability competitiveness.



Multiple MSP on Further Classes of Matroid

In our algorithms, incorporating an element $x \in \text{OPT}$ into the solution set ALG required *avoiding* certain elements. In transversal matroids, vertices sharing a partner with x were avoided. The need to identify such *blocking* elements highlights the relevance of the definition introduced in [3].

k -directed matroid. We define a k -directed matroid as any matroid that admits a k -directed certifier.

Matroid Class	Asymptotic Competitiveness
$(J, 1)$ -SP (1-Uniform) [2]	$1 - e^{-J/e(1+o(1))}$
r -Uniform	$1 - O(e^{-J/r})$
Transversal	$1 - e^{-J/e(1+o(1))}$
Graphic	$1 - o(e^{-J/2e})$
k -Directed	$1 - O(e^{-J/2k})$

Asymptotic competitiveness of the proposed algorithms.

This work was partially supported by FONDECYT Regular project 1231669 ‘Combinatorial Explorable Uncertainty’, CMM ANID Basal FB210005, and ANID National PhD Scholarship 21252789.

References

- [1] Bérczi, K., Livanos, V., Soto, J., y Verdugo, V., “Matroid secretary via labeling schemes”, 2024, <https://arxiv.org/abs/2411.12069> <https://arxiv.org/abs/2411.12069>.
- [2] Gilbert, J. P. y Mosteller, F., “Recognizing the maximum of a sequence”, Journal of the American Statistical Association, vol. 61, pp. 35–73, 1966, <https://api.semanticscholar.org/CorpusID:122946727> <https://api.semanticscholar.org/CorpusID:122946727>.
- [3] Marinkovic, J., Soto, J. A., y Verdugo, V., “Online combinatorial assignment innbsp;independence systems”, en Integer Programming and Combinatorial Optimization: 25th International Conference, IPCO 2024, Wroclaw, Poland, July 35, 2024, Proceedings, (Berlin, Heidelberg), p. 294308, Springer-Verlag, 2024.